

Assignment 2

Question 1: Trading Idea Generation

Overview: My trading idea is to implement a statistical arbitrage based on commodity-equity spread analysis. I've noticed that many companies and hence equities are heavily based on certain equities. For example, aircraft production companies, like Boeing, require large amounts of aluminum to build their aircrafts. My idea is to analyze the prices of raw material and commodity inputs that companies depend on to do business. The phenomena I hypothesize is that when the price of the commodity increases or decreases, this change may not be immediately priced into the company's equity, resulting in an opportunity for arbitrage. Such commodity-equity pairs are ubiquitous in our world. Based on my knowledge, other examples could be the cost of corn compared to packaged food stocks like those for corn chips, or the cost of natural gas and the equities related to fertilizer companies. Through extensive research, I plan to find pairs where an arbitrage strategy would be successful.

Logic: The basic logic is that companies need raw materials to operate and when those materials get more expensive, it costs the company more to produce their product, which would hurt their profits. The contrary would also be true when the raw material cost decreases. My idea hinges on the phenomena that the stock market does not price in these commodity price changes immediately, allowing for a window where the stock is mispriced relative to the change in commodity price.

Pattern: I will be studying commodity price movements and when they are significant. I plan to use statistical methodology here considering mean and standard deviation. I hypothesize that there would be a specific standard deviation level that would signify a large price movement upwards or downwards. At the same time, I would study relevant stock prices and learn how long it takes for the stock to "react." My hypothesis is that this would not happen immediately, but eventually the stock would catch up.

Model: I will start by finding pairs where the relationship between the commodity and stock is strong and direct. Initially, I would like to consider the three pairs (1) aluminum and aircraft

companies, (2) corn and packaged corn goods and (3) natural gas and fertilizer companies. I will do more research to understand if these pairs are strong and what other pairs may exist that have a better, direct relationship. For each pair, I would like to think about how much the stock has moved over a defined time period and how much the relative stock moved over that same period. I would essentially identify what that gap is against what the expected gap should have been. This gap is my key signal for arbitrage.

Trading Rules: When the commodity price increases by a significant amount, but the stock has not, I would short the stock. When the commodity price decreases significantly, but the stock has not, I take a long position in the relevant stock. If the gap closes and the stock catches up, I exit the trade. Alternatively, if the gap is not closing, but is rather increasing, going against expectation, I exit the trade to cut losses. The exact time period I would wait before exiting and the price change that constitutes a “significant” increase or decrease is dependent on analysis of the pairs and interactions.

Expected Outcomes: When the signal is right, the gap would close in the direction of the commodity prediction and I would close the trade profitably. When the signal is wrong, the gap would not close. This could happen because the company has already hedged on price corrections, or some other unrelated reason may impact the stock. Early on, I expect fewer pairs to work, until I learn the best method to select the pairs and statistical methodology to trade.

Challenges with other ideas I considered (alternative data):

1. PM 2.5 factory emissions to predict product stock. I started by considering PM 2.5 from a Rivian factory and found some correlation. When PM 2.5 emissions had peaked, there was a relative peak following in the Rivian stock as large production led to large sales. However, this was not scalable. Publicly available PM 2.5 sensor data is not always near factories and is easily polluted by other combusive emissions. This idea only worked for Rivian. When I considered tapping into carbon markets with this data for California Carbon Allowance (CCA) and Regional Greenhouse Gas Initiative (RGGI), I found that the combined PM 2.5 signal was not granular enough and would not cover the entire market of RGGI/CCA for a viable signal.

2. Social Media sentiment as a demand signal combined with statistical features of cryptocurrencies. Here, social media signals are not entirely free and publicly available data sources. When I tried to work with Reddit/Twitter, I hit limits fairly early in the process and did not engineer a strong signal. The time period of this course and resources I had available would not be feasible to continue down this path.

Question 2: Core Logic Design of Trading Strategies Table (Filled)

Core Logic Design of Trading Strategies		
Targeting Market	U.S. Equities: Publicly traded companies in commodity-intensive industries (industrials, consumer staples, agriculture, energy)	
Phenomena Description ☐	Theoretical Framework ☐	Phenomena: When a commodity input cost rises significantly, companies that rely on that input face margin compression, yet stock prices frequently do not adjust immediately. This lag, information diffusion lag, creates a pricing inefficiency. Framework: Efficient Market Hypothesis (semi-strong form) would suggest stock prices should immediately incorporate public commodity data. This strategy exploits the documented deviation from EMH, grounded in behavioral finance (investor inattention, analyst forecast stickiness).

Analytical Framework ☐	Theoretical Hypothesis ☐	Framework: Commodity-equity spread analysis. Calculate rolling cost-of-goods-sold (COGS) sensitivity (beta) of each equity to its primary input commodity. Compare realized commodity price change vs. implied stock price adjustment. Hypothesis: A statistically significant increase (or decrease) in a key input commodity price, which is not reflected in the dependent company's stock price within a defined lag window, predicts subsequent downward (upward) price correction in the stock. $Spread_t = (\beta \times \Delta C_t) - \Delta S_t$ Meaning, the Spread of mispricing gap at a given time t, is the difference of the change in stock (S) and product of the stock's sensitivity to the commodity moves (β) and the change in the commodity (C).
Phenomena Interpretation ☐	Theoretical Reasoning ☐	Interpretation: The lag represents investor inattention to cross-asset signals, analyst earnings estimate stickiness, and institutional delay in repricing. Equity markets are slower to process commodity market signals than commodity-forward markets. Reasoning: Companies cannot indefinitely absorb input cost increases without earnings erosion, nor will they forgo profit expansion when input costs fall. Fundamental value must eventually converge, creating a predictable directional trade.

Trading Opportunity	Cross-asset pricing gap between commodity futures/spot prices and the stock prices of companies with high commodity input dependency. Gap closes when analyst revisions, earnings surprises, or market repricing occurs (expected within 2–8 weeks of a significant commodity move).
Trading Operations	Long or short equity positions on identified stocks when commodity-equity spread exceeds a statistical threshold (relative to standard deviation). Enter on signal trigger; exit at mean-reversion target or stop-loss. No direct commodity position taken, rather this is purely equity play on input-cost mispricing.
Financial Instrument(s)	U.S.-listed equities (common stock).

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<https://claude.ai/chat/7e1d6cd3-ddba-48a0-9c79-286fd770d2cc>